

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - IX

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Class: TY.IT	Batch:
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

Student.cs

```
namespace practical_9A.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Age { get; set; }
        public string Course { get; set; }
    }
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using practical_9A.Models;
namespace StudentAPI.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }

        public DbSet<Student> Students { get; set; }
    }
}
```

appsettings.json

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"
  }
}
```

Program.cs

```
using Microsoft.EntityFrameworkCore;
using StudentAPI.Data;

var builder = WebApplication.CreateBuilder(args);
```

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```
builder.Services.AddControllers();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();

app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
```

```
app.Run();
```

StudentsController.cs

```
using Microsoft.AspNetCore.Mvc;
using practical_9A.Models;
using StudentAPI.Data;

namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;

        public StudentsController(AppDbContext context)
        {
            _context = context;
        }

        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }

        // GET BY ID
        [HttpGet("{id}")]
        public IActionResult GetStudent(int id)
        {
            var student = _context.Students.Find(id);

            if (student == null)
                return NotFound();

            return Ok(student);
        }

        // INSERT
        [HttpPost]
        public IActionResult AddStudent(Student student)
        {

```

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```
        _context.Students.Add(student);
        _context.SaveChanges();

        return Ok(student);
    }

    // UPDATE
    [HttpPut("{id}")]
    public IActionResult UpdateStudent(int id, Student student)
    {
        var data = _context.Students.Find(id);

        if (data == null)
            return NotFound();

        data.Name = student.Name;
        data.Age = student.Age;
        data.Course = student.Course;

        _context.SaveChanges();

        return Ok(data);
    }

    // DELETE
    [HttpDelete("{id}")]
    public IActionResult DeleteStudent(int id)
    {
        var student = _context.Students.Find(id);

        if (student == null)
            return NotFound();

        _context.Students.Remove(student);
        _context.SaveChanges();

        return Ok();
    }
}
```

practical_9A1.0OAS 3.0

<https://localhost:44337/swagger/v1/swagger.json>

Students

GET/api/Students

POST/api/Students

GET/api/Students/{id}

PUT/api/Students/{id}

DELETE/api/Students/{id}

WeatherForecast

GET/WeatherForecast

Schemas

Student >

WeatherForecast >

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POST /api/Students

Parameters Cancel Reset

No parameters

Request body application/json

Edit Value | Schema

```
{
  "name": "Omkar Sawant",
  "age": 20,
  "course": "Bsc.IT"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'https://localhost:44337/api/Students' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Omkar Sawant",
    "age": 20,
    "course": "Bsc.IT"
  }'
```

Request URL

PUT /api/Students/{id}

Parameters Cancel Reset

Name Description

id = required
integer(\$int32)
(path)

Request body application/json

Edit Value | Schema

```
{
  "name": "Omkar Sawant",
  "age": 20,
  "course": "Bsc.IT"
}
```

Execute Clear

Responses

Curl

```
curl -X 'PUT' \
  'https://localhost:44337/api/Students/2' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Omkar Sawant",
    "age": 20,
    "course": "Bsc.IT"
  }'
```

SQL Server Object Explorer

- SQL Server
 - (localdb)\MSSQLLocalDB (SQL Server 17.0.4025 - IT0)
 - Databases
 - System Databases
 - EmployeeDB
 - ProductDB
 - StudentDB
 - Tables
 - System Tables
 - External Tables
 - Dropped Ledger Tables
 - dbo._EFMigrationsHistory
 - dbo.Students**
 - Views

dbo.Students [Data] | dbo.Students [Design] | StudentsController.cs | Program.cs

Max Rows: 1000

	Id	Name	Age	Course
	1	Omkar Sawant	20	Bsc.IT
	2	Omkar Sawant	20	Bsc.IT
*	NULL	NULL	NULL	NULL